



ENGF0034: Design and Professional Skills

Introductory lecture

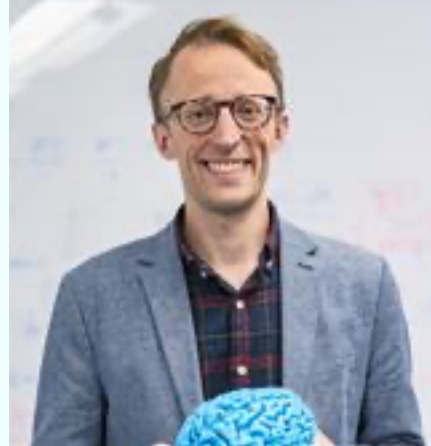
Prof. Martin Benning
06 October 2026



ENGF0034 2026/27 is a team effort



Charlotte Maughan
Jones



James Cole



Vivek Ramachandran



Amy Lourenço



Sunny Bains

Teaching Assistants



Cris Tudosie



Anya Lakhani



Shafiyyah Shafiyyah

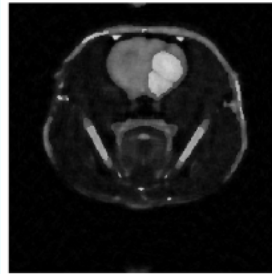
Who am I? Professor of Inverse Problems



Martin Benning

Areas of expertise:

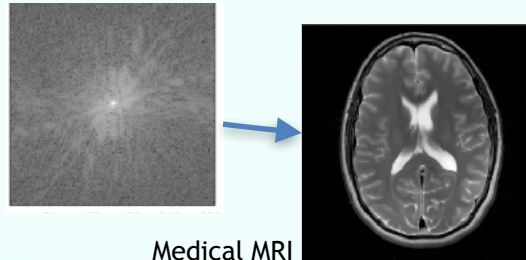
- Inverse & Ill-posed problems
- Regularisation theory
- Machine learning
- Optimisation
- Industrial & medical Imaging
- Image processing



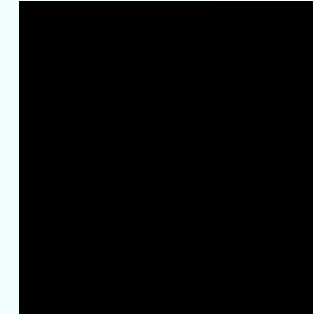
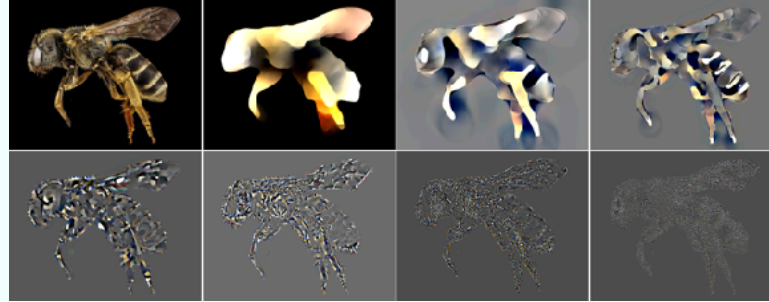
Joint reconstruction
& segmentation



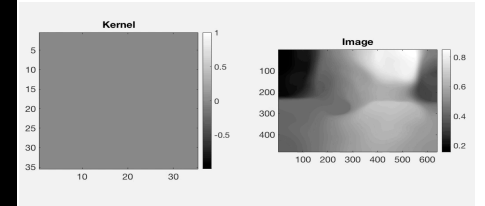
Nonlinear spectral decomposition



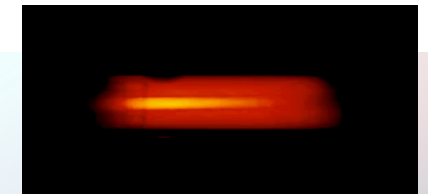
Medical MRI



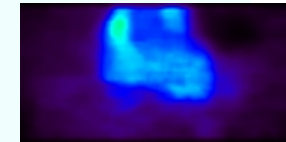
Blind image deconvolution



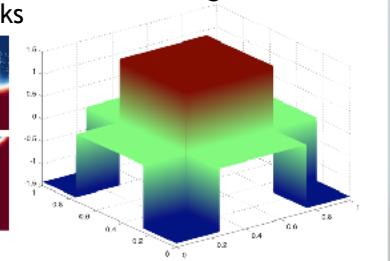
Industrial MRI



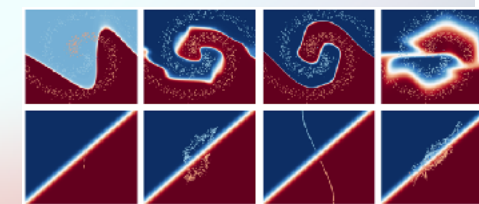
Medical PET



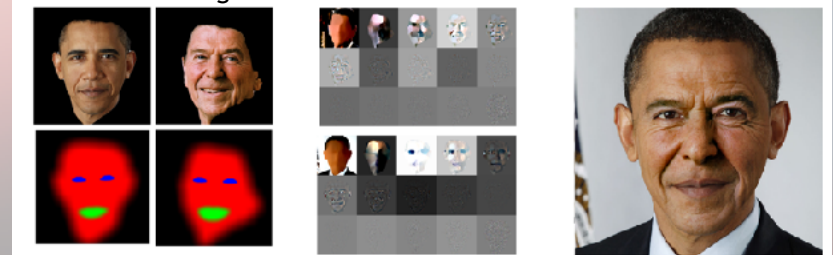
Nonlinear eigenfunctions



Classification with residual neural networks



Nonlinear image fusion



Transmission Electron Microscopy / Tomography

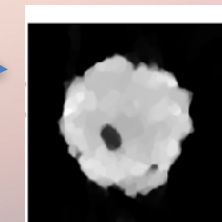
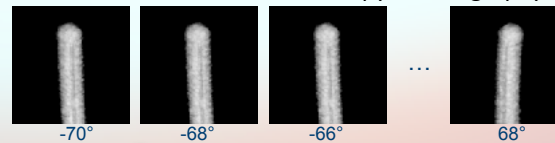
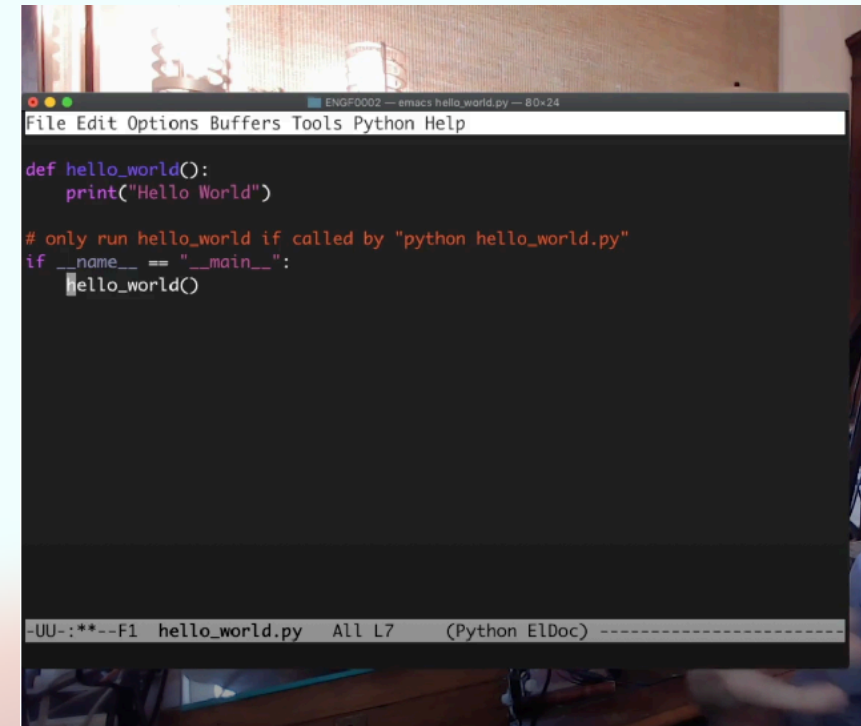


Image reconstruction

Module approach

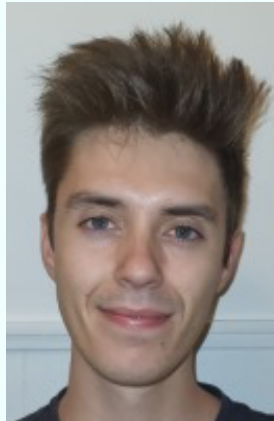
- Lectures (almost) every **Tuesday**
- Tutorials (almost) every **Friday**
- **Ask and answer questions** via the [Moodle forum](#)
- **Course materials** will be released via [git](#)
- **Assessments** will be released via git or Moodle
- **Drop-in clinics** will take place as extra support for those new to programming



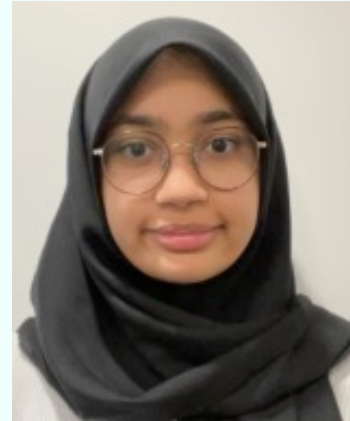
Let us checkout the module [Moodle](#) & [git](#) pages

Drop-in clinics

Teaching Assistants



Cris Tudosie



Shafiyyah Shafiyyah



Anya Lakhani

- Drop-in clinic Cris (in-person): Wednesday, 3-4 pm
G03, Henry Morley
 - Drop-in clinic Shafiyyah (in-person): Thursday, 4-5 pm
G03, Henry Morley
 - Drop-in clinic Anya (in-person): Friday 1-2 pm
G03, Henry Morley
- (Starts next week!)

Assignments

- Week 1 - 2: Introductory programming assignment
- Week 2 - 3: Debug the bomber game
- Week 3 - 5: Debug the frogger game
- Week 5 - 7: Write a tetris autoplayer (20%)
- Week 7 - 9: Write pacman protocol specification (15%)
- Week 7 in Term 2: Ethics report (15%)
- Term 2 Reading Week: Presentations (10%)
- + Scenario assessments in Term 2 (remaining 40%)

Assignments

- Week 1 - 2: Introductory programming assignment
- Week 2 - 3: Debug the bomber game
- Week 3 - 5: Debug the frogger game
- Week 5 - 7: Write a tetris autoplayer (20%)
- Week 7 - 9: Write pacman protocol specification (15%)
- Week 7 in Term 2: Ethics report (15%)
- Term 2 Reading Week: Presentations (10%)
- + Scenario assessments in Term 2

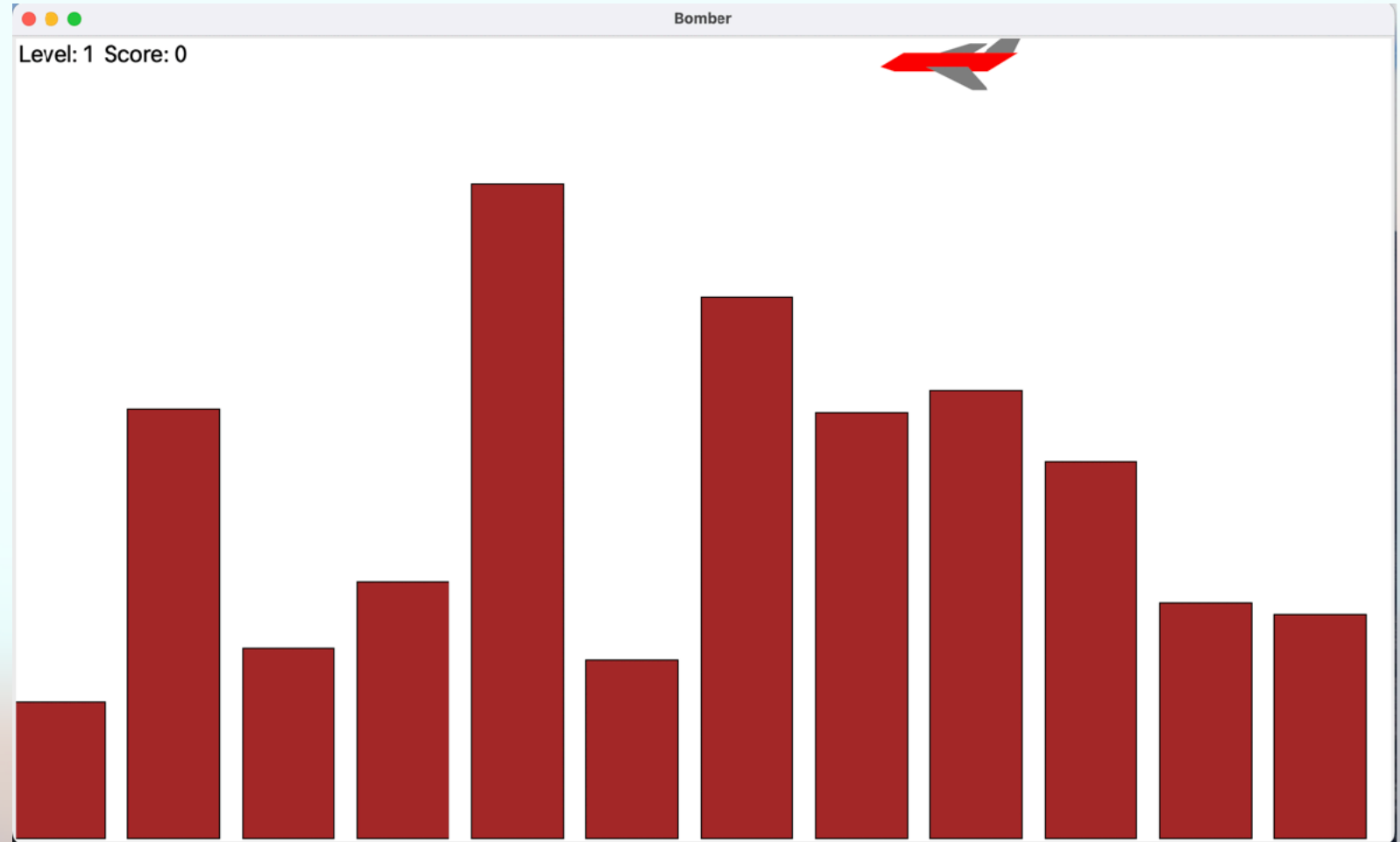
Do not plan time away
during reading week!

Assignments

Assignment 2: (Canyon) Bomber

Classic Atari Arcade Game from 1977

Task: Find five bugs that make the game unplayable, and fix them!

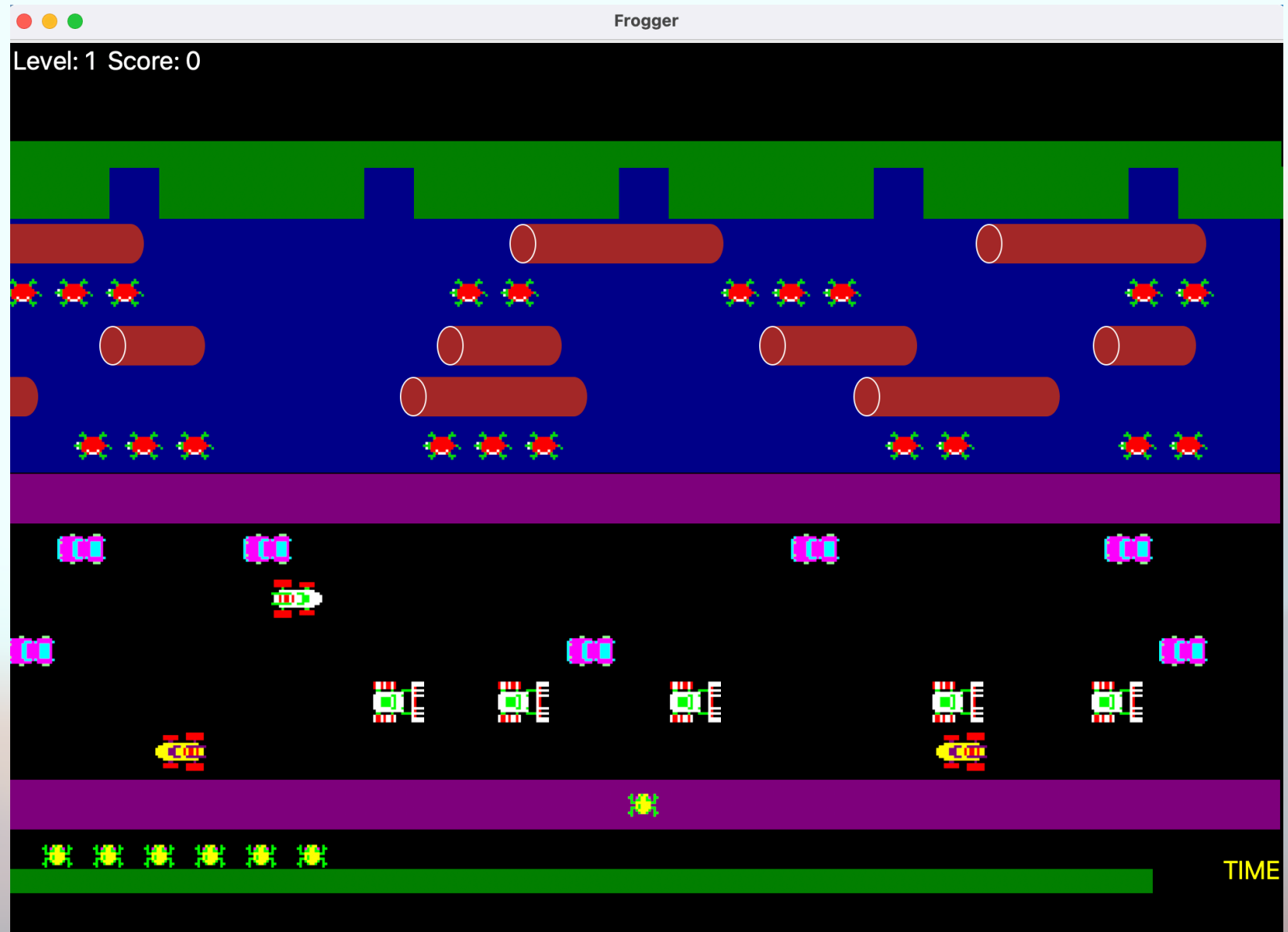


Assignments

Assignment 3: Frogger

Classic Konami/Sega Arcade
Game from 1981

Task: Find five bugs that make
the game unplayable, and fix
them!



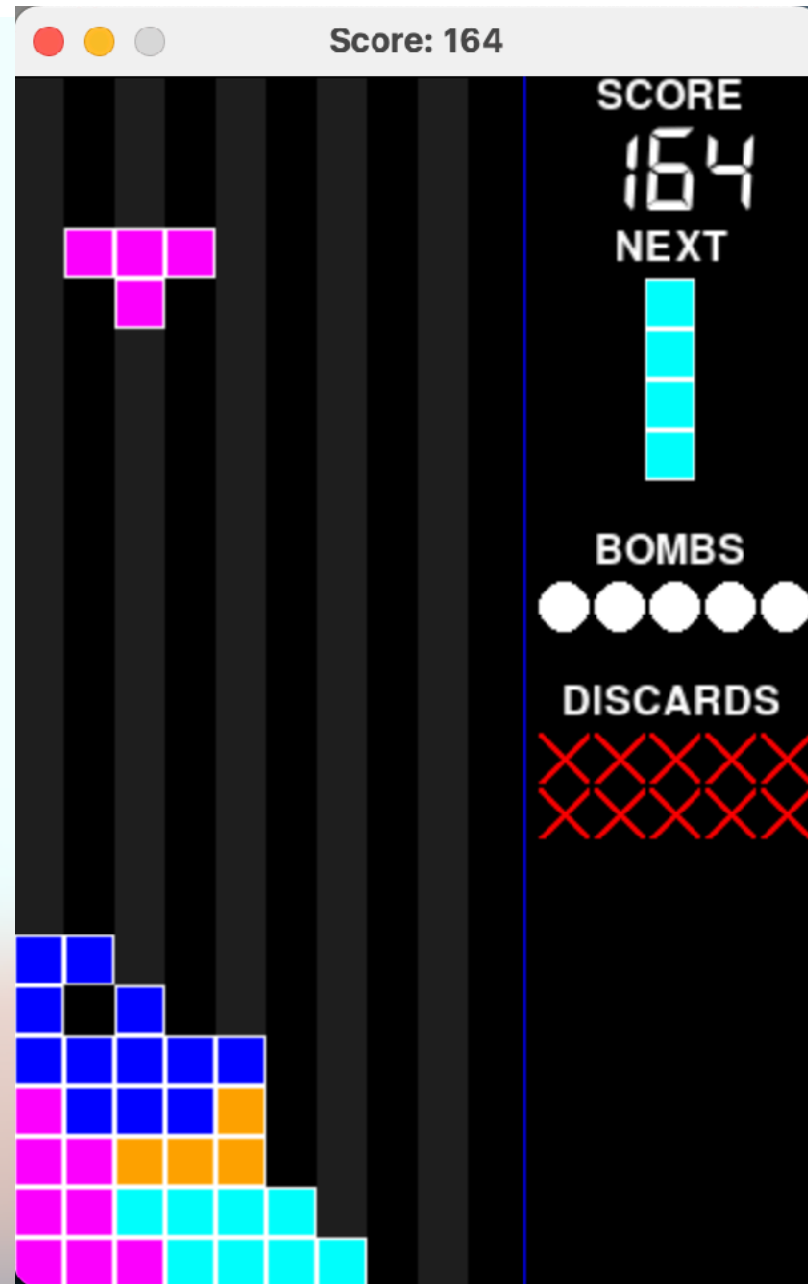
Assignments

Assignment 4: Tetris autoplayer

Puzzle video game created by
Alexey Pajitnov in 1985

Game Boy version by Nintendo
popularised Tetris in 1989

Task: write a program that
plays Tetris and compete with
your peers



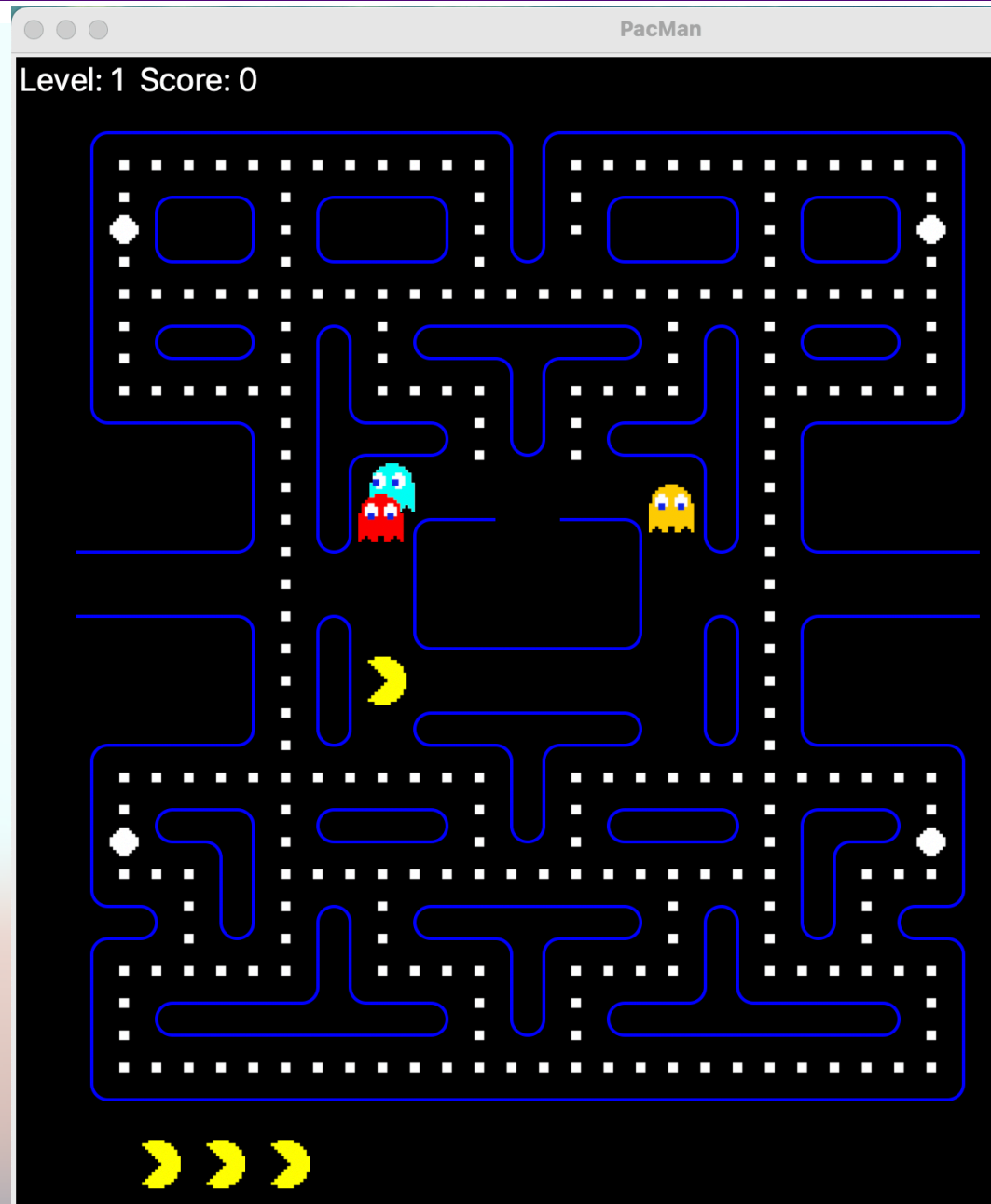
Assignments

Assignment 5: Write specification for Pacman game

1980 Maze video game

Special network version of the game

Task: write a better
network protocol
specification for this game

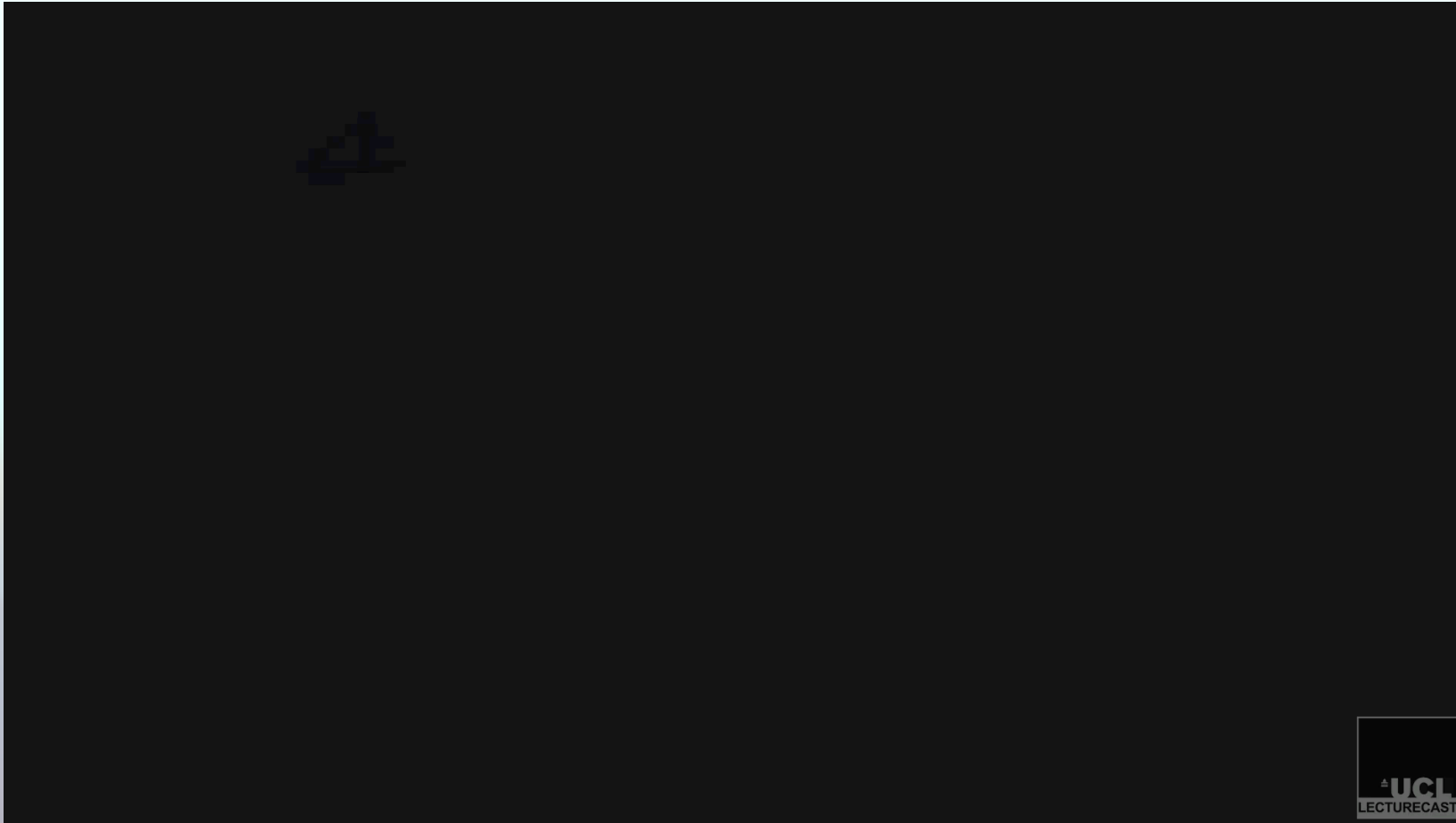


Other remarks

- For assessments 1 - 3 communicate and work with your peers
- If you have questions, it is likely your peers will have similar questions ➡ ask via Moodle forum, **not** via e-mail
- Do not only use forum when you have questions; help your peers and answer their questions, too!
- However, do not ask or post solutions for summative (marked) assessments (4 & 5) ➡ punishable offence
- Make use of forum and the drop-in clinics; do not suffer in silence!
- I know it is tempting to use Large Language Models (like ChatGPT, Gemini, Claude etc.), but there are risks:
 - Often you require programming experience to understand LLM outputs
 - More importantly, you require experience to identify **wrong** outputs!
 - It is not allowed to use LLMs for summative (marked) assessments!

This Friday (October 9)

Pebble in the pond!



When? 11 am - 1 pm

Where?

Royal National Hotel

Next Tuesday (October 12)

Teamwork - Live Session

with Dr Charlotte
Maughan Jones



When? 11 am - 1 pm

Where?

TBA

Now / Next hour ➡ Programming foundations!